WORKED SOLUTION

STEP III 2023, Question 4

Pure • Chebyshev Polynomial Coefficients

STEP III 2023, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Writing $\cos((2n+1)\theta)$ as the real part of $(\cos\theta + \mathrm{i}\sin\theta)^{2n+1}$ turns p into an explicit sum, and each coefficient of p becomes a sum of binomial coefficients. Sums of the form $\sum\binom{N}{2r}$ or $\sum r\binom{N}{2r}$ that run over even indices only are found by putting $t=1$ and $t=-1$ into $(1+t)^N$ (or its derivative) and adding or subtracting, because the odd-index terms then cancel. For the final part we compare the top few coefficients on both sides of the factorisation.

Solution

Throughout, write $c = \cos\theta$ and $s = \sin\theta$.

Part (i)

By de Moivre's theorem and the binomial theorem,

$$\cos((2n+1)\theta) + \mathrm{i}\sin((2n+1)\theta) = (c+\mathrm{i}s)^{2n+1} = \sum_{k=0}^{2n+1}\binom{2n+1}{k}c^{2n+1-k}(\mathrm{i}s)^k.$$

Since c and s are real, the real terms are exactly those with k even, say $k=2r$ with $0 \le r \le n$, and then $(\mathrm{i}s)^{2r} = (-1)^r s^{2r} = (-s^2)^r = (c^2-1)^r$. Taking real parts,

$$\cos((2n+1)\theta) = \sum_{r=0}^{n}\binom{2n+1}{2r}\cos^{2n+1-2r}\theta\,(\cos^2\theta - 1)^r,$$

as required.

Part (ii)

Part (i) shows that

$$\mathrm{p}(x) = 1 + \sum_{r=0}^{n}\binom{2n+1}{2r}x^{2n+1-2r}(x^2-1)^r$$

for every $x = \cos\theta \in [-1, 1]$. Both sides are polynomials that agree at infinitely many points, so they are the same polynomial.

The rth term has degree $(2n+1-2r)+2r = 2n+1$ with leading coefficient $\binom{2n+1}{2r}$, and the constant 1 does not affect x^{2n+1}. So the coefficient of x^{2n+1} in p is

$$S_0 = \sum_{r=0}^{n} \binom{2n+1}{2r},$$

the sum of the even-index binomial coefficients. Putting $t = 1$ and $t = -1$ into $(1+t)^{2n+1} = \sum_k \binom{2n+1}{k} t^k$ gives

$$2^{2n+1} = \sum_k \binom{2n+1}{k}, \qquad 0 = \sum_k (-1)^k \binom{2n+1}{k}.$$

Adding, the odd-k terms cancel and the even-k terms double: $2S_0 = 2^{2n+1}$. Hence the coefficient of x^{2n+1} in $\mathrm{p}(x)$ is $S_0 = 2^{2n}$.

Part (iii)

In the rth term, $x^{2n+1-2r}(x^2-1)^r$, we need the x^{2r-2} term of $(x^2-1)^r$ to reach x^{2n-1}. That term is $\binom{r}{1}(x^2)^{r-1}(-1) = -rx^{2r-2}$ (and it is absent when $r = 0$, consistent with $-r = 0$). Since $n \geq 1$ we have $2n-1 \geq 1$, so the constant 1 in p plays no part. The coefficient of x^{2n-1} is therefore

$$-\sum_{r=0}^{n} r \binom{2n+1}{2r}.$$

Ask yourself How can we pick out the even-index terms when each one is weighted by r?

Differentiating brings down the index as a factor. From $(1+t)^{2n+1} = \sum_k \binom{2n+1}{k} t^k$,

$$(2n+1)(1+t)^{2n} = \sum_{k=1}^{2n+1} k \binom{2n+1}{k} t^{k-1}.$$

At $t = 1$ this gives $\sum_k k\binom{2n+1}{k} = (2n+1)2^{2n}$. At $t = -1$ the left-hand side is 0 (since $2n \geq 1$), so

$$0 = \sum_k (-1)^{k-1} k \binom{2n+1}{k} = \sum_{k \text{ odd}} k \binom{2n+1}{k} - \sum_{k \text{ even}} k \binom{2n+1}{k}.$$

So the even-k and odd-k parts are equal, and each is half the total:

$$\sum_{k \text{ even}} k \binom{2n+1}{k} = (2n+1)2^{2n-1}.$$

With $k = 2r$, this says $\sum_r 2r\binom{2n+1}{2r} = (2n+1)2^{2n-1}$, that is, $\sum_r r\binom{2n+1}{2r} = (2n+1)2^{2n-2}$. Hence the coefficient of x^{2n-1} in $\mathrm{p}(x)$ is $-(2n+1)2^{2n-2}$, as required.

Part (iv)

The leading coefficient. By (ii), p has degree $2n+1$ with leading coefficient 2^{2n}. If q has degree m and leading coefficient a, then $(x+1)[\mathrm{q}(x)]^2$ has degree $2m+1$ and leading coefficient a^2. So $m=n$ and $a^2=2^{2n}$, and since $a>0$, the coefficient of x^n in q is 2^n.

One more fact about p. Every term $x^{2n+1-2r}(x^2-1)^r$ contains only odd powers of x, because $x^{2n+1-2r}$ is an odd power and $(x^2-1)^r$ has only even powers. The only even power in p is the constant 1. As $2n\geq 2$, the coefficient of x^{2n} in p is 0.

Comparing coefficients. Now take $n\geq 2$ and write

$$\mathrm{q}(x)=2^n x^n+bx^{n-1}+cx^{n-2}+\cdots,$$

so that

$$[\mathrm{q}(x)]^2=2^{2n}x^{2n}+2^{n+1}b\,x^{2n-1}+(b^2+2^{n+1}c)x^{2n-2}+\cdots.$$

Multiplying by $x+1$, the coefficients of x^{2n} and x^{2n-1} in $(x+1)[\mathrm{q}(x)]^2$ are

$$2^{n+1}b+2^{2n}\qquad\text{and}\qquad(b^2+2^{n+1}c)+2^{n+1}b.$$

The coefficient of x^{2n} in p is 0, so $b=-2^{2n}/2^{n+1}=-2^{n-1}$. Then $b^2=2^{2n-2}$ and $2^{n+1}b=-2^{2n}$, and matching the coefficient of x^{2n-1} with part (iii) gives

$$2^{2n-2}+2^{n+1}c-2^{2n}=-(2n+1)2^{2n-2}.$$

Rearranging,

$$2^{n+1}c=2^{2n-2}(-(2n+1)-1+4)=2^{2n-2}(2-2n)=2^{2n-1}(1-n),$$

so the coefficient of x^{n-2} in q is

$$c=2^{n-2}(1-n),$$

as required.

Checks

- For $n=1$: $\cos 3\theta=4\cos^3\theta-3\cos\theta$, so $\mathrm{p}(x)=4x^3-3x+1=(x+1)(2x-1)^2$. The leading coefficient is 2^2, the x-coefficient is $-3=-3\cdot 2^0$, and q has leading coefficient 2^1.
- For $n=2$: $\mathrm{p}(x)=16x^5-20x^3+5x+1=(x+1)(4x^2-2x-1)^2$. The x^3-coefficient is $-20=-5\cdot 2^2$, and the constant term of q is $-1=2^0(1-2)$.
- For $n=3$ we found $\mathrm{q}(x)=8x^3-4x^2-4x+1$, with coefficient of x equal to $-4=2^1(1-3)$. A computer check confirmed every coefficient formula for $n=1,\ldots,12$ by expanding $\cos\big((2n+1)\theta\big)+1$ exactly and taking the square root of $\mathrm{p}(x)/(x+1)$.

Common slips

- Forgetting the x^{2n} coefficient of p. It is 0, and that is exactly what fixes the x^{n-1} coefficient b of q, which you need before you can find c.
- Writing $(\mathrm{i}s)^{2r} = s^{2r}$ and losing the sign $(-1)^r$. That sign is what produces $\cos^2\theta - 1$ rather than $1 - \cos^2\theta$.
- Putting only $t = 1$ into the derivative in (iii), which gives the sum over all k, not just the even ones.

Beyond STEP

The polynomial T_N with $T_N(\cos\theta) = \cos N\theta$ is the Chebyshev polynomial of degree N, and parts (ii) and (iii) find its top two coefficients. The factorisation in (iv) reflects the fact that $T_{2n+1}(x) + 1$ has a simple root at $x = -1$ and a double root at each other point where $\cos\big((2n+1)\theta\big) = -1$. Chebyshev polynomials are central in approximation theory: among monic polynomials of degree N, $2^{1-N}T_N$ is the one with the smallest maximum size on $[-1, 1]$. Siklos, *Advanced Problems in Mathematics*, has more questions in this vein.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.